

DATASET

a. Sample Audio Files Used

The system was tested using a variety of audio sources to ensure robust performance across different sonic profiles. The primary dataset includes:

- **Original Input:** nhac.wav (A high-quality, lossless multi-instrumental music track).
- **Pre-processed Stems (via Audacity):** The original track was separated into four distinct categories for individual compression:
 - vocals.wav: Lead and backing vocal tracks.
 - drums.wav: Percussive elements (kick, snare, hi-hats).
 - bass.wav: Low-frequency melodic elements.
 - other.wav: Synthesizers, guitars, and background textures.
- **Processed Outputs:** Compressed versions of each stem (e.g., vocalscompress.wav, basscompress.wav) used for the final reconstruction.

b. Data Labeling & Characteristics

The inputs are labeled based on their acoustic characteristics to evaluate the compression algorithm's impact on specific frequency ranges:

- **Speech/Vocals:** Labeled as Vocals. Used to test intelligibility and the preservation of harmonic clarity.
- **Music (Multi-channel/Mixed):** Labeled as nhac.wav. Used as the baseline for the "original mix" vs. "reconstructed mix" comparison.
- **Low-Frequency (Bass):** Labeled as Bass. Used to analyze how well the system maintains low-end "thump" under 200 Hz.
- **Transient-Heavy (Drums):** Labeled as Drums. Used to detect "smearing" or loss of punch in percussive transients.

c. Instructions to Obtain Dataset

To reproduce the dataset used in this project, follow these steps:

1. **Source Acquisition:** Place any standard lossless .wav file (16-bit or 24-bit, 44.1kHz) into the project root folder.
2. **Manual Separation (Audacity):**

- Open **Audacity** and import your source audio.
 - Use the "Split New" or separation tools to isolate the four categories: *Vocals, Drums, Bass, and Other*.
 - Export these as individual .wav files into the output/stems/ directory.
3. **Automated Separation (Optional):** If using the demucs library included in the requirements, run the separation script to automatically generate the four stems.
4. **Compression Parameters:** Apply FFmpeg encoding to the stems using bitrates ranging from **32 kbps to 320 kbps** to generate the test inputs for the reconstruction phase.

File:

[nhac.wav](#)